

Session - 9

18/09/2026

* Inequalities - 8

1) $R \geq S \geq F, F \leq 0 = U, U \geq V \geq W$

- a) $R > F$ F b) $R = F$ F c) $U \geq W$ T
d) $S > 0$ F e) $S \leq V$ F

a & b - either or

2) $A \geq B \geq C = D \geq E < F > G$

- a) $A > C$ F either or b) $B > C$ F either or
 $A = C$ F $B = D$ F
c) $C > E$ F either or d) $E > G$ F Neither nor
 $D = E$ F $E = G$ F

Imp * Non-Relation either or

9) $A \geq B \geq C < D$

- 1) $A > C$ F } either or
2) $A \neq C$ F } (Relational)
 $A \geq C$ T

→ No relation (i) ✓

- 1) $A \geq D$ F } (ii) ✓ (iii) variables same ✓
2) $A \leq D$ F } (iv) follows (iv) rule also ✓
 ← No relation.

either or (Non-Relational)

1) $A \geq D$ rule (iv) No follows ie Neither nor

2) $A \geq D$

1) $A > D$ rule (iv) no follows ie Neither nor

2) $A < D$

eg- 1) $P > Q < R \geq J > L < M$

a) $L > R$

b) $R \geq L$

Rule (i) no follows

Neither nor ✓

2) $M \geq D < N < P < R$

a) $N < R$ (F)

b) $R \geq M$ (F)

Rule (i) ✓

(ii) ✓

Neither nor ✓

(iii) ✓

(iv) X

3) $A = B \geq C \geq D < E$

a) $A \geq E$ (F)

b) $E = A$ (F)

Rule (i) ✓

(ii) ✓

Neither nor

(iii) ✓

(iv) X

4) $A > M, A \leq B, R < S \leq B$

a) $A \geq R$ F

b) $R > A$ F

Rule (i) ✓

Non relational

(ii) ✓

either or

(iii) ✓

(iv) ✓

5) $A = B \geq C \geq D < E$

a) $D = A$ F

b) $B > D$ F

Rule (i) X

neither nor

either or (Relational)

• Direction (Q.11-15):

- $A \$ B$ A is not smaller than B. $\geq$
 $A \# B$ A is not greater than B. $\leq$
 $A @ B$ A is neither smaller than nor equal to B. $>$
 $A \wedge B$ A is neither smaller than nor greater than B. $=$
 $A \cdot B$ A is neither greater than nor equal to B. $<$

11) $H < J$, $J = N$, $N > R$

- a) $R < J$ FF b) $H > J$ F c) $N > H$ T

12) $M > J$, $J \geq T$, $T = N$

- a) $N \leq J$ T b) $T < M$ T c) $M > N$ T

13) $D = K$, $K \leq F$, $F > P$

- a) $P > D$ F b) $K \leq P$ F c) $F \geq D$ T

14) $K \leq N$, $N \geq T$, $T < J$

- a) $J > N$ F b) $K > T$ F c) $T > K$ F

15) $M > D$, $D = V$, $V \geq W$

- a) $W > M$ F b) $M < V$ FF c) $D \geq W$ T

• Directions (Q.16-20):

- $A @ B$ A is not smaller than B. $\geq$
 $A \# B$ A is neither smaller than nor equal to B. $>$
 $A \cdot B$ A is not greater than B. $\leq$
 $A \$ B$ A is neither greater than nor equal to B. $<$
 $A * B$ A is neither smaller than nor greater than B. $=$

16) $J > H, H = T, T < R, R \leq F$

a) $J > R$ F b) $R > F$ F c) $J > T$ T

17) $E < P, P \leq H, H \geq I, I > K$

a) $P = I$ F b) $I \leq E$ F c) $H \leq K$ F

18) $L \geq K, K > R, R < H, H \leq N$

a) $L > R$ T b) $N > R$ T c) $L = N$ F

19) $T < V, V \leq K, K = L, L \leq H$

a) $V \leq H$ T b) $T < L$ T c) $T < H$ T

20) $V \geq W, W > D, D < M, M \leq F$

a) $V > D$ T b) $F > D$ T c) $V = F$ F

Directions (Q. 21-25):

$P \wedge Q$	P is not smaller than Q $\geq$
$P * Q$	P is neither greater than nor smaller than Q $=$
$P @ Q$	P is neither greater than nor equal to Q $<$
$P \$ Q$	P is not greater than Q $\leq$
$P \# Q$	P is neither smaller than nor equal to Q $>$

21) $M < T, T \leq R, R \geq J$

a) $J > M$ F b) $R > M$ T c) $J = T$ F

22) $D \geq B, B > H, H = F$

a) $F < B$ T b) $F < D$ T c) $H < D$ T

23) $H = M, M < T, T \leq K$

a) $K > M$ T b) $T > H$ T c) $H < K$ T

24) $R = T, T < M, M \leq K$

a) $K < R$ F

b) $M > R$ T

c) $K > T$ T

25) $F > W, W \leq M, M \geq R$

a) $R \leq W$ F

b) $F > R$ F

c) $W = R$ F

Imp • Directions (Q. 26-30):

$P \wedge Q$ P is either greater than or equal to Q $\geq$

$P \vee Q$ P is either smaller than or equal to Q $\leq$

$P @ Q$ P is neither smaller than nor greater than Q. $=$

$P \$ Q$ P is smaller than Q $<$

$P \delta Q$ P is greater than Q $>$

26) $H \geq T, T \leq M, M > F$

a) $F < T$ F

b) $H > M$ F

27) $B = N, N \geq T, T < K$

a) $T = B$ F

b) $T < B$ F (either I or II)

28) $R < J, J > F, F \leq H$

a) $H > J$ F

b) $R < F$ F

29) $J > P, P = N, N \leq F$

a) $J > F$ F

b) $F \geq P$ T

30) $B > T, T < H, H = M$

a) $M > T$ T

b) $B > H$ F

• Directions (Q. 31-35):

$P \cdot Q$	P is neither greater than nor equal to Q . $<$
$P * Q$	P is neither smaller than nor equal to Q . $>$
$P @ Q$	P is not greater than Q . $\leq$
$P \# Q$	P is not smaller than Q . $\geq$
$P \wedge Q$	P is neither greater than nor smaller than Q . $=$

31) $D > K$, $K < N$, $N \leq F$

a) $D > N$ F b) $F > N$ F c) $D \leq N$ F

Only either I or III is true

32) $H \geq R$, $R \leq M$, $M = N$

a) $H > M$ (F) b) $N > R$ F c) $N = R$ F

only either II or III is true

33) $M \leq R$, $R < T$, $T = N$

a) $N > R$ (T) b) $T > M$ (T) c) $N \geq R$ (F)

Only I and II are true

34) $B \geq R$, $R > K$, $K < M$

a) $M > R$ F b) $K < B$ T c) $K \leq B$ F

only II is true

35) $F < D$, $D > K$, $K \leq B$

a) $B > D$ F b) $F < K$ F c) $B > F$ F

Direction (Q. 36-40):

$P \wedge Q$	P is not smaller than Q	$\geq$
$P * Q$	P is not greater than Q	$\leq$
$P @ Q$	P is neither greater than nor equal to Q	$<$
$P \$ Q$	P is neither smaller than nor equal to Q	$>$
$P \% Q$	P is neither greater than nor smaller than Q	$=$

36) $J > K$, $K \leq T$, $T < N$, $N \geq R$

- a) $J > T$ F b) $R \leq T$ F
c) $N > T$ T d) $R \leq K$ F

Only III is true

37) $F \geq W$, $W \geq R$, $R < M$, $M > D$

- a) $D < R$ F b) $M > R$ F
c) $R < D$ F d) $R \leq F$ T

Only IV is true

38) $H < B$, $B \leq E$, $V \geq E$, $W > V$

- a) $W > E$ T b) $H < E$ T
c) $H < V$ F d) $W > B$ T

All are true

39) $R \geq K$, $K \leq N$, $N < J$, $J = H$

- a) $R > N$ F b) $J < K$ F
c) $H < N$ T d) $R > H$ F

Only III is true

- 40) $K \leq D, D > N, N = M, M \geq W$
- a) $M < K$ F b) $N < K$ F
c) $M < D$ T d) $W \leq N$ T

Only II and IV are true

Session-10

21/09/2026

* Inequality-9

- Definitely False:

$$P \geq Q \geq R$$

- 1) $P > R$ (F) 2) $P < R$ (Definitely false) 3) $P \geq R$ (T) 4) $P \leq R$ (F) 5) $P = R$ (F)

$\{ >, <, \geq \} \rightarrow$ from True Conclusion
Take this for definitely false

$$A > B \geq C$$

- 1) $A > C$ (F) 2) $A < C$ (F) 3) $A \geq C$ (F) 4) $A \leq C$ (F) 5) $A = C$ (F)

IMP $\rightarrow$ If there is no relation betⁿ the variables then there will be No definitely false

- Directions (Q. 79-81): For the given statement, find the conclusion that definitely does not follow.

Statement - $A > B \geq C = D$

- 79) 1) $A < D$ (DF) 2) $A \leq D$ (DF) 3) $A \geq D$ (T)
4) $A \geq D$ 5) $A = D$ (DF)

- 80) 1) $B < D$ (DF) 2) $B \leq D$ 3) $B > D$
4) $B \geq D$ (T) 5) $B = D$

- 81) 1) $C < D$ (D.F.) 2) $C \leq D$ 3) $C > D$ (D.F.)
4) $C \geq D$ 5) $C = D$ (T)